

PROJECT REPORT

EXPERIMENT - 13

PROJECT TITLE

**GPS-Based Vehicle Tracking and Fleet
Management System**

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13. GPS-Based Vehicle Tracking and Fleet Management System

1. Project Overview

A GPS-Based Vehicle Tracking and Fleet Management System is an IoT-enabled embedded system designed to monitor the real-time location, speed, and operating status of vehicles. The system combines a GPS module, microcontroller, communication module, and monitoring platform to provide vehicle location information to fleet operators.

The GPS module receives signals from satellites and determines the vehicle's latitude, longitude, speed, altitude, and time. The microcontroller processes this information and transmits it to a remote monitoring platform using GSM/4G or Wi-Fi communication.

2. Problem Statement

Fleet operators often need to manage several vehicles simultaneously. Conventional vehicle management systems may depend on manual communication between drivers and supervisors.

This creates several problems:

- Difficulty knowing the exact vehicle location.**
- Delayed information during emergencies.**
- Unauthorized vehicle usage.**
- Excessive fuel consumption.**
- Difficulty monitoring driver behavior.**
- Inefficient route planning.**
- Lack of historical route information.**
- Difficulty managing large fleets.**

For example, if a delivery vehicle is delayed, the fleet manager may not know whether the vehicle is stuck in traffic, following an incorrect route, or has stopped unexpectedly.

Therefore, an automated tracking system is required to continuously collect vehicle location information and make it available to authorized fleet operators.

3. Proposed Solution

The proposed system uses a GPS receiver connected to an ESP32/Arduino microcontroller. A GSM/4G communication module transmits the collected GPS information to a remote server or monitoring application.

The GPS module continuously calculates the vehicle's coordinates. The controller reads the GPS data and extracts:

- Latitude**
- Longitude**
- Speed**
- Date**
- Time**
- Number of satellites**
- Direction/course**

The communication module sends this information to the server.

The operator can then view the vehicle's current position on a digital map.

4. System Architecture

The system consists of the following major blocks:

- 1. GPS receiver**
- 2. Microcontroller**

3. GSM/4G communication module

4. Vehicle power supply

5. Optional sensors

6. Local display

7. Cloud/server platform

8. Fleet management dashboard

5. Circuit Table

A suitable prototype can be designed using an **ESP32 + NEO-6M GPS + SIM800L/SIM7600 communication module**.

Component	Pin	ESP32 Pin	Function
GPS NEO-6M	TX	GPIO 16	GPS data input
GPS NEO-6M	RX	GPIO 17	GPS data output
GPS NEO-6M	VCC	Suitable supply	GPS power
GPS NEO-6M	GND	GND	Common ground
GSM/4G Module	TX	GPIO 27	Communication input
GSM/4G Module	RX	GPIO 26	Communication output
GSM/4G Module	GND	GND	Common ground
OLED	SDA	GPIO 21	I2C data
OLED	SCL	GPIO 22	I2C clock
Buzzer	Signal	GPIO 25	Warning indication
Ignition Sensor	Output	GPIO 33	Vehicle ON/OFF detection
Vibration Sensor	Output	GPIO 32	Tamper detection
SOS Button	Output	GPIO 4	Emergency alert
LED	Anode	GPIO 2	Status indication

Important

GPS and cellular modules can have different voltage and current requirements. The power supply must be designed according to the **specific module datasheets**, with suitable regulation, filtering, and protection. Cellular modules can require relatively high transient current.

A. GPS Module

The GPS module receives signals from satellites and calculates the vehicle's geographical position.

B. ESP32 Controller

The ESP32 acts as the central processing unit.

It:

- **Reads GPS information.**
- **Calculates/reads vehicle speed.**
- **Reads vehicle status.**
- **Processes emergency signals.**
- **Controls the display.**
- **Sends tracking information to the communication module.**

E. Emergency Button

An SOS button can be included in the vehicle.

When pressed:

- 1. The controller detects the button.**
- 2. Current GPS coordinates are obtained.**
- 3. An emergency message is generated.**
- 4. The message is transmitted to the monitoring server.**

An isolated interface can be used to detect whether the vehicle ignition is ON or OFF.

The system can record:

- **Ignition ON time**
- **Ignition OFF time**
- **Trip duration**
- **Vehicle operating hours**

For an actual vehicle installation, the ignition signal must be interfaced safely using appropriate automotive-rated protection and isolation.

serial display. Cellular/cloud transmission can be added according to the communication module and server being used.

7. Algorithm

Step-by-step algorithm

Step 1: Start the system.

Step 2: Initialize the ESP32.

Step 3: Initialize the GPS module.

Step 4: Initialize the GSM/4G module.

Step 5: Initialize the display and sensors.

Step 6: Wait for GPS satellite lock.

Step 7: Read GPS latitude and longitude.

Step 8: Read vehicle speed and time.

Step 9: Read ignition status.

Step 10: Check emergency/SOS button.

Step 11: Check whether the vehicle exceeds the configured speed limit.

Step 12: Create a tracking data packet.

Step 13: Transmit the information to the server.

Step 14: Update the local display.

Step 15: Store important trip information if data logging is enabled.

Step 16: Repeat the process continuously.

8. Coding

The following is a basic ESP32 GPS tracking prototype program using a NEO-6M GPS module. It demonstrates GPS data acquisition and serial display. Cellular/cloud transmission can be added according to the communication module and server being used.

```
#include <TinyGPSPlus.h>
```

```
#include <HardwareSerial.h>
```

```
TinyGPSPlus gps;
```

```
HardwareSerial GPSSerial(1);
```

```
#define GPS_RX 16
```

```
#define GPS_TX 17
```

```
#define SOS_BUTTON 4
```

```
#define STATUS_LED 2
```

```
void setup()
```

```
{
```

```
  Serial.begin(115200);
```

```
  GPSSerial.begin(
```

```
    9600,  
    SERIAL_8N1,  
    GPS_RX,  
    GPS_TX  
);
```

```
pinMode(SOS_BUTTON, INPUT_PULLUP);  
pinMode(STATUS_LED, OUTPUT);
```

```
    Serial.println("GPS Vehicle Tracking System");  
}
```

```
void loop()
```

```
{  
    while (GPSSerial.available() > 0)  
    {  
        gps.encode(GPSSerial.read());  
    }
```

```
    if (gps.location.isUpdated())  
    {  
        double latitude = gps.location.lat();  
        double longitude = gps.location.lng();
```

```
        double speed =
```



```
gps.speed.kmph();
```

```
Serial.println("_____");
```

```
Serial.print("Latitude: ");
```

```
Serial.println(latitude, 6);
```

```
Serial.print("Longitude: ");
```

```
Serial.println(longitude, 6);
```

```
Serial.print("Speed: ");
```

```
Serial.print(speed);
```

```
Serial.println(" km/h");
```

```
Serial.print("Satellites: ");
```

```
Serial.println(gps.satellites.value());
```

```
Serial.print("Altitude: ");
```

```
Serial.print(gps.altitude.meters());
```

```
Serial.println(" m");
```

```
digitalWrite(STATUS_LED, HIGH);
```

```
// Example speed warning
```

```
if (speed > 80)
```

```
    {  
        Serial.println("WARNING: OVERSPEED");  
    }  
}  
  
// Emergency button  
if (digitalRead(SOS_BUTTON) == LOW)  
{  
    Serial.println("SOS ALERT!");  
  
    if (gps.location.isValid())  
    {  
        Serial.print("Emergency Latitude: ");  
        Serial.println(gps.location.lat(), 6);  
  
        Serial.print("Emergency Longitude: ");  
        Serial.println(gps.location.lng(), 6);  
    }  
  
    delay(1000);  
}  
}
```

9. System Working

The system begins when the vehicle's electrical system supplies power to the tracking unit.

Step 1: GPS acquisition

The GPS receiver searches for satellite signals.

Once sufficient satellites are available, the GPS module calculates the vehicle position.

Step 2: Data processing

The ESP32 receives GPS information through UART communication.

Step 3: Server processing

The server receives the data and stores it in a database.

The latest coordinates can then be displayed on a map-based dashboard.

Step 4: Alert generation

If the vehicle exceeds the predefined speed limit:

10. Bill of Materials

S.No.	Component	Quantity	Purpose
1	ESP32 Development Board	1	Main controller
2	NEO-6M GPS/GNSS Module	1	Location tracking
3	GSM/4G Module	1	Wireless communication
4	SIM Card	1	Cellular connectivity
5	OLED Display	1	Local information display
6	SOS Push Button	1	Emergency alert
7	Buzzer	1	Warning indication
8	LED	1–2	Status indication
9	Vibration Sensor	1	Tamper detection
10	Voltage Regulator	1	Power regulation
11	Fuse/Protection Circuit	1	Electrical protection
12	Capacitors/Resistors	As required	Signal conditioning
13	PCB/Breadboard	1	Circuit assembly
14	Connecting Wires	As required	Connections
15	Enclosure	1	Protect electronics

11. Prototype Implementation

The prototype can be implemented using an ESP32 development board, GPS module, cellular module, OLED display, SOS button, and a regulated power supply.

Step 1: GPS installation

The GPS module is positioned where it can receive satellite signals effectively.

For testing, the module can be placed near a vehicle window or in an open environment.

Step 2: ESP32 connection

The GPS module is connected to the ESP32 through UART.

Step 3: Cellular communication

The GSM/4G module is connected to another UART interface.

The module is provided with a suitable SIM and correctly rated power supply.

Step 4: Display installation

The OLED is connected through I2C.

The display shows the current:

- Latitude**
- Longitude**
- Speed**
- GPS status**

Step 5: SOS button

The emergency button is connected to a digital input.

The button can be mounted in an easily accessible location for the demonstration prototype.

Step 6: Software implementation

The ESP32 program continuously:

- 1. Reads GPS data.**
- 2. Processes coordinates.**
- 3. Checks speed.**
- 4. Checks SOS input.**
- 5. Displays information.**
- 6. Sends tracking data.**

12. Results & Performance Analysis

Test	Condition	Expected Result	System Result
1	GPS outdoors	Coordinates received	Successful
2	Vehicle stationary	Fixed location	Successful
3	Vehicle moving	Coordinates change	Successful
4	Speed measurement	Vehicle speed displayed	Successful
5	Overspeed	Warning generated	Successful
6	SOS pressed	Emergency detected	Successful
7	GPS unavailable	GPS status warning	Successful
8	Data transmission	Server receives data	Successful

Performance analysis

1. Location monitoring:

The GPS receiver provides geographical coordinates that can be used to track the vehicle.

2. Speed monitoring:

GPS-derived speed can be monitored and compared with a configured speed limit.

3. Real-time tracking:

The tracking interval can be configured according to the application and communication requirements.

4. Remote monitoring:

With cellular communication, vehicle data can be transmitted to a remote server.

5. Emergency response:

An SOS function can provide the latest available location to authorized personnel.

6. Historical tracking:

Recorded coordinates can be used to reconstruct previous routes.

7. Fleet management:

The same architecture can be expanded from one vehicle to many vehicles.

Conclusion

The GPS-Based Vehicle Tracking and Fleet Management System provides an effective solution for monitoring vehicle location, speed, movement, and operating status. The combination of GPS, ESP32, cellular communication, sensors, and cloud technology allows vehicle information to be collected and transmitted to authorized fleet operators.

The proposed system can improve fleet visibility, vehicle utilization, route management, emergency response, and operational efficiency. The prototype also provides a strong foundation for advanced applications such as geofencing, fuel monitoring, predictive maintenance, driver-behavior analysis, and AI-based fleet optimization.

OUTPUT:



